Huaiyuan Yao

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Google Scholar: 209 citations, h-index 8 (Nov. 2026)

Education

Arizona State University, U.S. Fall 2025 – present

Ph.D. in Computer Science, SCAI. Advisor: Prof. Hua Wei (DaRL Lab). Expected graduation: Spring 2029. Research interest: AI agents that keep learning from their own experience after real-world deployment.

Xi'an Jiaotong University, China 2021 – 2025

B.S. in Computer Science. Advisor: Prof. Yuanqi Su.

Research Experience

Honda Research Institute USA, Mountain View, CA May – Aug 2026

Research Intern (mentor: Manish Saroya). LLM-steered deterministic driving planners (HarnessDriver).

DaRL Lab, Arizona State University 2024 – present

Graduate Research Associate (2025 – present); *Research Intern*, remote (2024 – 2025). CoMAL, Instructional Agents, self-evolving multi-agent LLMs (LangMARL, MASkills), and test-time training.

Institute for AI Industry Research (AIR), Tsinghua University, China 2024 – 2025

Research Assistant (advisors: Prof. Peng Li, Prof. Yilun Chen). LiloDriver: lifelong motion planning.

Publications (7 first-authored)

Peer-reviewed conference papers

[1] **Huaiyuan Yao**, Longchao Da, Xiaoou Liu, Charles Fleming, Tianlong Chen, Hua Wei. *LangMARL: Natural Language Multi-Agent Reinforcement Learning.* EMNLP 2026 (Main).

[2] **Huaiyuan Yao**, Xiaoou Liu, Charles Fleming, Tianlong Chen, Hua Wei. *MASkills: Continual Skills Optimization for Multi-Agent LLM Systems.* Findings of EMNLP 2026.

[3] **Huaiyuan Yao**, Wanpeng Xu, Justin Turnau, Nadia Kellam, Hua Wei. *Instructional Agents: Reducing Teaching Faculty Workload through Multi-Agent Instructional Design.* EACL 2026. GitHub: 500 stars.

[4] **Huaiyuan Yao**, Zhenxiao Cao, Zhongman Wang, Hongyi Cai, Yuqiang Han, Xiao Luo. *How Effective is In-Context Learning with Large Language Models for Rare Cell Identification in Single-Cell Expression Data?* IEEE ICDM 2025.

[5] **Huaiyuan Yao**, Longchao Da, Vishnu Nandam, Justin Turnau, Zhiwei Liu, Linsey Pang, Hua Wei. *CoMAL: Collaborative Multi-Agent Large Language Models for Mixed-Autonomy Traffic.* SDM 2025, **Best Poster**.

[6] Tiejin Chen, Maunil Rohitbhai Vyas, **Huaiyuan Yao**, Hua Wei. *Steering Vectors as a Training Signal in LLM Post-Training.* NeurIPS 2026.

[7] Tiejin Chen, **Huaiyuan Yao**, Jia Chen, Evangelos E. Papalexakis, Hua Wei. *Every Response Counts: Quantifying Uncertainty of LLM-based Multi-Agent Systems through Tensor Decomposition.* ACL 2026.

[8] Yihan Hong, **Huaiyuan Yao**, Bolin Shen, Wanpeng Xu, Hua Wei, Yushun Dong. *RULERS: Locked Rubrics and Evidence-Anchored Scoring for Robust LLM Evaluation.* EMNLP 2026 (Main).

[9] Ziqi Wang, **Huaiyuan Yao**, Hua Wei, Hang Zhou, Xiao Luo. *AURORA: Attention-Intended Dual Verification with Adaptive Rollback for LLM Agents.* IROS 2026.

[10] Jacob Igo, **Huaiyuan Yao**, Wanpeng Xu, Nadia Kellam, Hua Wei. *From Course Concept to Lecture Video: An AI-Powered System for Automated MOOC Development.* LERN 2026.

Preprints and papers under review

[11] **Huaiyuan Yao**, David Isele, Sangjae Bae, Hua Wei, Manish Saroya. *HarnessDriver: Steering Deterministic Driving Planners toward Adaptive Closed-Loop Behavior.* Under review at ICRA 2027.

[12] **Huaiyuan Yao**, Pengfei Li, Bu Jin, Yupeng Zheng, An Liu, Lisen Mu, Qing Su, Qian Zhang, Yilun Chen, Peng Li. *LiloDriver: A Lifelong Learning Framework for Closed-Loop Motion Planning in Long-Tail Autonomous Driving Scenarios.* arXiv:2505.17209, 2025.

[13] Longchao Da, Tiejin Chen, Zhuoheng Li, Shreyas Bachiraju, **Huaiyuan Yao**, Li Li, Yushun Dong, Xiyang Hu, Zhengzhong Tu, Dongjie Wang, Yue Zhao, Ben Zhou, Ram Pendyala, Benjamin Stabler, Yezhou Yang, Xuesong Zhou, Hua Wei. *Generative AI in Transportation Planning: A Survey.* arXiv:2503.07158, 2025.

[14] Lina Zhao, Jiaxing Bai, Zihao Bian, Qingyue Chen, Yafang Li, Guangbo Li, Min He, **Huaiyuan Yao**†, Zongjiu Zhang. *Autonomous Multi-Modal LLM Agents for Treatment Planning in Focused Ultrasound Ablation Surgery.* Under review at IEEE JBHI. († corresponding author)

[15] Ferdous Al Rafi, Susrik Mukherjee, Latika Liladhar Dekate, Jennifer Yawa Lavoe, **Huaiyuan Yao**, Shlok Mohanty, Longchao Da, Xuesong Zhou, Hua Wei. *Sim2Signal: Sim-to-Real Benchmarks for Traffic Signal Control.* Under review at AAAI 2027.

[16] Wenwen Li, Pingping Wang, Yuanyuan Tian, Sizhe Wang, **Huaiyuan Yao**, Longchao Da, Al Mehdi Saadat Chowdhury, Mahdi Ahmadi, Yi-Wen Mo, Ferdous Al Rafi, Hua Wei, YooJung Choi, Paulo Shakarian. *LifelineQA: An Evidence-Grounded Benchmark for Situational-Awareness Question Answering in Emergency Operations.* Under review at ICLR 2027.

[17] Qingyue Chen, **Huaiyuan Yao**, Wanpeng Xu, Varun Shourie, Qianwen Wang, Hua Wei. *MAVE: A Multi-Agent Framework for Grounded Pedagogical Feedback from Online Class Videos.* arXiv:2610.24498, 2026.

★ Grants, Fellowships, and Honors

NSF ACCESS allocation, Principal Investigator: test-time training for self-improving LLM agents	2026
Google TPU Research Cloud (TRC): TPU compute support	2026
Best Poster Award, SIAM International Conference on Data Mining (SDM), for CoMAL	2025
University Graduate Fellowship, Arizona State University	Spring and Summer 2026
Global Security Initiative (GSI) Fellowship, Arizona State University	Fall 2025
Selected participant, AI-SCORE 2026 workshop on OR and CS, UIUC	2026
Finalist Prize, Mathematical Contest in Modeling (MCM), COMAP	2023

🏛 Service, Teaching, and Mentoring

- **Reviewer:** AAAI 2027 Student Abstract and Poster Program (PC), ECML-PKDD 2026, RLC 2026, IROS 2026, ACML 2026, WSDM 2026, EWRL 2026, CIKM 2026, RV 2026, ICMLA 2026, PAKDD 2026.
- **Community:** Volunteer, China3DV 2025; Founder, Neural Modeling and Programming Workshop (2024), a student-run computational neuroscience workshop with hundreds of participants.
- **Teaching Assistant, ASU:** CSE 598 Agentic AI (Fall 2026); CSE 475 Foundations of Machine Learning (Fall 2025); CSE 220 Programming for Computer Engineering (Spring 2026), with Prof. Robert K. Atkinson.
- **Mentoring:** undergraduate and master's students Jacob Igo, Siyuan Xing, Aarshdeep Nandra, Yuxuan Huang, Rayan Kazi (ASU), Ziqi Wang (Rutgers), Iris Hong (WashU), Qingyue Chen (Wuhan University).
- **Open source:** Instructional Agents (500 GitHub stars), CoMAL, LiloDriver, LangMARL, MASkills.